Shuqi Ke

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Employment

Shenzhen Shita Robotics Technology Co., Ltd. *Shenzhen, China*
Algorithm Engineer Intern *Mar. 2026 - Aug. 2026*

- Built a real-robot visual-tactile data collection, recording, training, and inference pipeline for embodied intelligence and edge vision; integrated Piper leader-follower control, LTeach tactile cameras, and multimodal data management; adapted ACT and PI0 / PI0.5 training configurations and effective batch-size accounting.
- In parallel, maintained a visual-tactile optical-flow estimation SDK, optimized hardware solutions, and deployed algorithms; developed two-frame motion-vector prediction, multi-camera detection, and template tracking; implemented FP16 / PTQ INT8 quantization, calibration-set construction, and PC/RV1126B/RK3588 consistency validation with GStreamer/MIPI pipelines, concurrent processing, profiling, and on-site calibration.

Education

Jiujiang University *Jiujiang, China*
IoT Engineering *Sep. 2022 - Jun. 2026*

- **Core Coursework**: Data Structures, Computer Networks, Operating Systems, Computer Organization, Advanced Programming.
- **Language Certifications**: CET-4 and CET-6.

Research Experience

CVI Lab, Jiujiang University
Undergraduate Research Assistant · Computer Vision *Dec. 2022 - Sep. 2025*

- Conducted independent and collaborative research in object detection, medical image segmentation, and edge AI; handled data annotation and processing; investigated ResNet and DenseNet backbones and segmentation models such as SAM; designed experiments, implemented models, performed comparative, ablation, and error analyses; and contributed to wire-sequence model research, manuscript preparation, and experimental work for a first-author paper.
- Developed and maintained a PyQt5-based end-to-end biomedical imaging workflow for chromosome preprocessing, candidate-region detection, result review, abnormal-sample annotation, batch loading, manual correction, and structured result export; responsible for UI development and related front-end/back-end maintenance.

Research Interests

Tactile perception, multimodal robot learning, and algorithm optimization for contact-rich manipulation, spanning visual-tactile data collection, multimodal policy training, tactile representation and force-field modeling, sim-to-real transfer, and edge-device deployment; progressively exploring world models for embodied intelligence.

Publications

[1] TY-YOLO: An Attention-Based Multi-Scale Complex Scene Fire Smoke Detection Algorithm [Under Review]
First Author · Pattern Recognition Letters

- Led model selection, architecture optimization, loss-function tuning, and manuscript preparation; combined attention mechanisms with multi-scale feature fusion to address small-target smoke, occlusion, and false positives in complex backgrounds.

[2] FgFEU-Net: A Fine-Grained Feature Extraction U-Net Model for Breast Tumor Segmentation Based on Transfer Learning [Paper]
Breast Ultrasound Segmentation · Co-author *EDGE 2025 · Springer LNCS 16154 (2026)*

- Developed FgFEU-Net, a transfer-learning U-Net integrating a VGG16 encoder, same-level feature fusion, and Combo Loss to improve tumor-boundary delineation and mitigate foreground–background imbalance in breast ultrasound; achieved 99.03% accuracy, 96.83% precision, and 99.35% sensitivity on BUS2018.

[3] The Real-Time Intelligent Transportation Detection System Based on Edge-Cloud Collaborative Computing [Paper]
Edge–Cloud Intelligent Transportation · Co-author *EDGE 2025 · Springer LNCS 16154 (2026)*

- Deployed YOLOv8n on edge devices for vehicle recognition, traffic-flow analysis, congestion monitoring, wrong-way driving detection, and boundary-crossing alerts; integrated edge–cloud collaboration to offload centralized processing, achieving 55.06 FPS and an F1 score of 0.84 on edge hardware.

Project Portfolio

VTLA *LeRobot · PI0.5 · Piper*

VLA / Robot Learning

- Built a LeRobot 0.5.2-based visual-tactile learning extension stack for real-world robots while preserving upstream compatibility; integrated AgileX Piper leader-follower arms, LTeach tactile sensing, and data collection, training, and evaluation workflows.
- Extended ACT and PI0/PI0.5 with tactile inputs and implemented DreamTacVLA: a torque-matrix-driven Think–Dream–Act branch on the ACT backbone with a unified data contract for robot state, RGB observations, tactile force fields, and action sequences.

TactiWeave-VP *PyTorch · ResNet-18 · Causal TCN*

Visual-Physical Tactile Fusion

- Built a fabric-classification research prototype for short tactile interactions by jointly modeling tactile-pressure image sequences and calibrated six-dimensional physical states: tri-axial force, planar resultant, friction coefficient, and contact area.
- Combined a single-channel ResNet-18 visual encoder with a physical-state MLP, followed by frame-wise fusion, dilated causal TCN processing, and attention pooling for temporal classification.

IsaacSim-Tactile4OpenWorld *Isaac Lab · UIPC · Force Fields*

Tactile Simulation & Force Fields

- Built a tactile-contact research platform in Isaac Sim/Isaac Lab using UIPC/libuipc to model rigid-deformable contact, compliant tactile-membrane deformation, and local Fx/Fy/Fz force-field reconstruction.
- Unified tactile RGB, force fields, robot assets, and tasks in one workflow for Piper and Franka grasping, lifting, insertion, and rolling experiments, producing RGB, metadata, and HDF5 research data.

OpenFireAlert *YOLO11 · TensorRT · Jetson*

Wildfire Monitoring & Alerting

- Built an end-to-end edge-intelligence platform for wildfire and fire/smoke monitoring, integrating YOLO11/TensorRT inference on Jetson, ESP32-S3 sensor nodes, and a Spring Boot/MySQL control plane with a web console.
- Unified camera, temperature/humidity, smoke, and buzzer signals in one alert pipeline; used per-device debouncing and cooldown mechanisms to coordinate buzzer, email, and dashboard alerts.

TactileFlowField *PyTorch · RKNN · RV1126B*

Visual-Tactile Dense Optical Flow

- Built a dense optical-flow learning and edge-deployment system for visual-tactile sensors, covering multi-version model management, training, PyTorch streaming inference, and ONNX/RKNN conversion.
- Predicted dense optical flow from the current frame and an initial reference frame; supported FP16 and PTQ INT8 inference, plus RV1126B multi-stream sequential inference, ROI processing, baseline calibration, and device management.

EdgeVisTrack-RKNN *RKNN · Cython · RV1126B*

Sparse Multi-Camera Marker Tracking

- Built a pure-vision multi-camera sparse circular-marker tracker for RV1126B: regional RGB thresholding, morphological closing, and Hough circles establish the first frame; local ROI template correlation, subpixel localization, and loss recovery track subsequent frames.
- Provided CPU, Fixed RKNN, and Dynamic RKNN backends. Fixed uses embedded shared low-rank templates, while Dynamic retains each marker's first-frame appearance through runtime coefficients.

Vision Workbench *PySide6 · OpenCV · Release Engineering*

Computer Vision Workbench

- Built a unified PySide6 desktop application exposing seven computer-vision workflows through modular Python APIs, from classical image processing and panorama reconstruction to YOLO detection, segmentation, and training.
- Productized the application with optional dependency isolation, 184 automated test functions, cross-platform Qt/accessibility checks, and verified wheel/Windows EXE updates using version gates, SHA-256 validation, dependency fingerprints, and rollback.

AdaptiveUI-SKILL *Python · Static Analysis · Agent Skills*

Responsive UI Audit & Repair

- Designed two explicit-invocation Agent Skills that turn responsive UI work into an audit-repair-verification workflow across HTML/CSS, React, Vue, Svelte, and Tailwind projects.
- Implemented a zero-dependency Python analyzer with 24 stable rules, severity/confidence metadata, suppressions, JSON Schema output, and CI thresholds, backed by 79 automated tests.

AgentTools *PowerShell · Diagnostics · Codex*

Agent Engineering & Diagnostics

- Built a read-only Codex diagnostics tool covering PowerShell profiles, Conda hooks, Python encoding, Git line endings, proxy/TUN state, and reconnect logs, without exposing credentials or modifying the system.
- Added cross-platform AGENTS.md templates, task handoff, and runtime Skill inventory for explicit-invocation plugins.

Skills

Robot	PyTorch, LeRobot, ACT, PI0/PI0.5, Piper, Visual-Tactile Sensing
Vision	OpenCV, YOLO11, U-Net
Systems	GStreamer, RTSP, MIPI, HDF5, Qt (PySide6/PyQt5), Spring Boot 3, ESP32-S3
Simulation	Isaac Sim, Isaac Lab, UIPC/libuipc
Programming	Python, Cython, Java, C/C++
Edge Deployment	ONNX, TensorRT, RKNN/RKNNLite, FP16, PTQ INT8, Jetson, RV1126B, RK3588